Industrial Internship Report on

” Banking Information System ”

Prepared by

[Shiyam C]

Executive Summary
This report describes the work completed during the Industrial Internship organized by Upskill Campus in collaboration with UniConverge Technologies Pvt Ltd (UCT). The internship duration was six weeks and aimed to provide practical exposure to real-world software development and problem-solving. The project developed during this internship is a Banking Information System using Java. The system simulates basic banking operations such as account creation, deposit, withdrawal, balance inquiry, and account details management. The project was implemented using object-oriented programming concepts in Java. The system provides a simple console-based interface that allows users to interact with the banking services. Through this project, I gained practical experience in programming, system design, and software development. This internship helped me improve my technical knowledge and problem-solving skills.

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1 Preface

Industrial internships are essential for students to gain practical exposure to real-world applications of their academic knowledge. This six-week internship program provided an opportunity to work on a project that simulates a banking system.

The project assigned was **Banking Information System**, which focuses on implementing basic banking operations using Java programming. The system allows users to create accounts, deposit and withdraw money, check balance, and view account details.

This internship helped me understand software development practices and improve my programming skills.

I would like to thank **Upskill Campus**, **UniConverge Technologies Pvt Ltd**, and the mentors who guided me throughout the internship. Their support and guidance helped me successfully complete this project.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

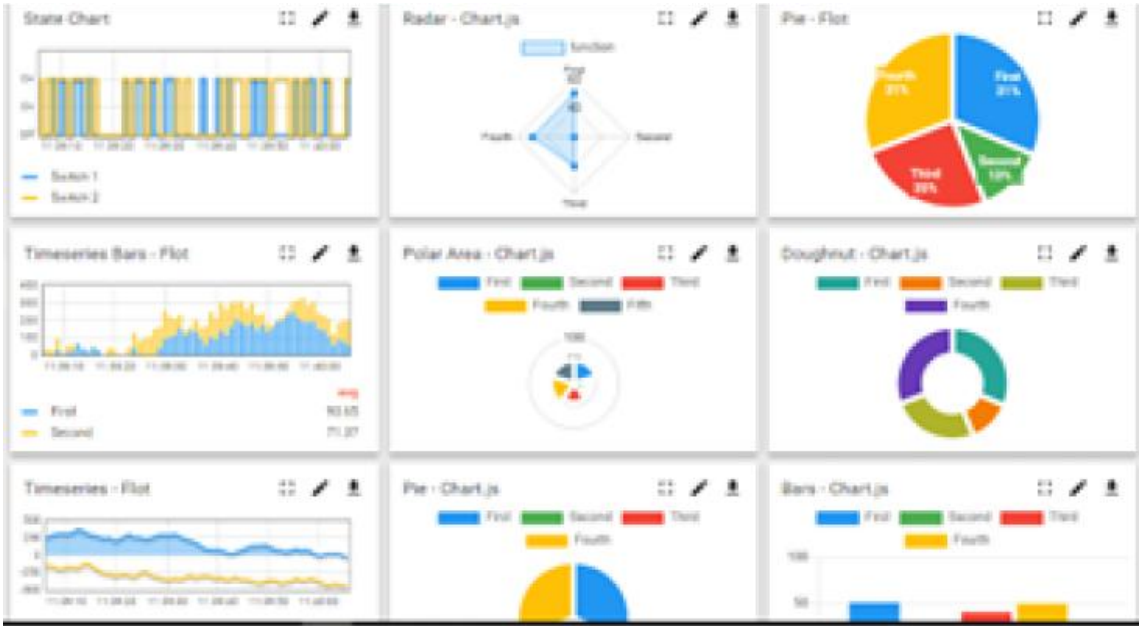
i. UCT IoT Platform (uct Insight)

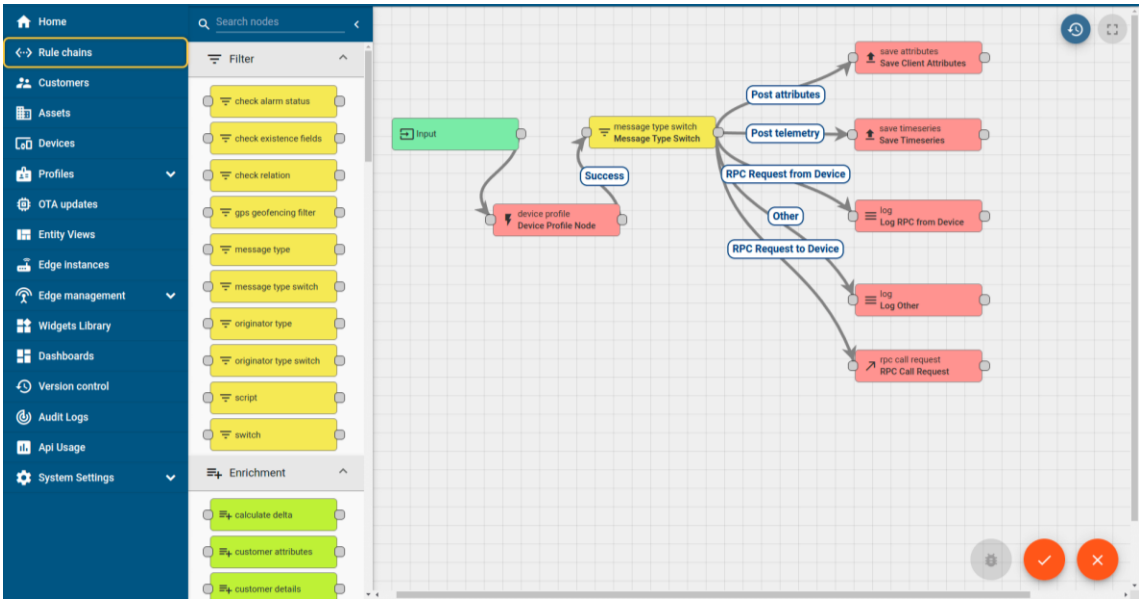
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

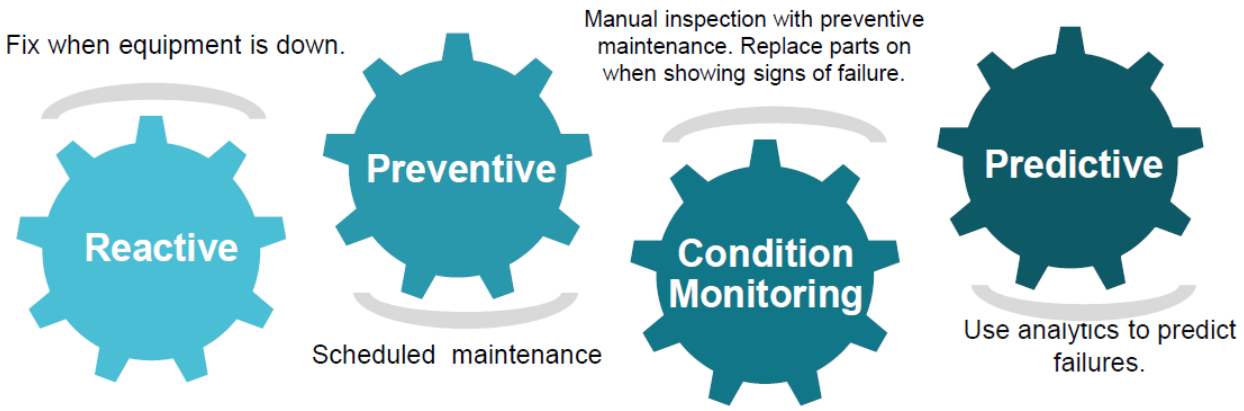
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

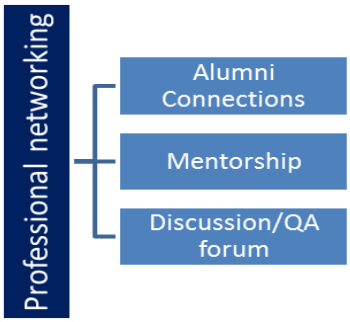
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

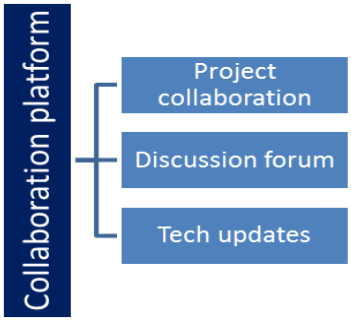


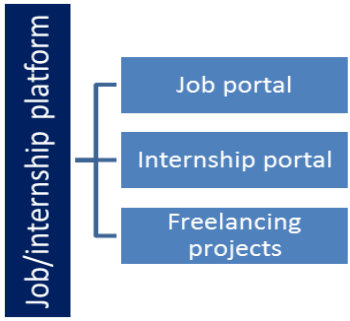
2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Java Programming Documentation – https://docs.oracle.com

[2] GeeksforGeeks Java Programming Articles – https://www.geeksforgeeks.org/java/

[3] Visual Studio Code Documentation – https://code.visualstudio.com/docs

2.6 Glossary

Terms	Acronym
OOP	Object-Oriented Programming – A programming paradigm based on objects and classes
Java	A high-level programming language used for building applications
Class	A blueprint used to create objects in programming
Object	An instance of a class
Account	A bank record containing user details and balance

3.Problem Statement

Banks require systems to manage customer accounts and financial transactions efficiently. Traditional manual banking systems can lead to errors and inefficiencies in handling financial records.

The objective of this project is to develop a **Banking Information System** that allows users to perform basic banking operations such as:

- Creating bank accounts
- Depositing money
- Withdrawing money
- Checking account balance
- Viewing account details

The system aims to simulate basic banking functionality using Java programming.

4 Existing and Proposed solution

Existing System:

Traditional banking systems relied on manual record keeping or outdated software systems. These systems had several limitations:

- Time-consuming manual processes
- Possibility of human errors
- Lack of automation
- Limited accessibility

Proposed System:

The proposed system is a Java-based Banking Information System that automates basic banking operations.

The system provides the following features:

- Account creation
- Deposit and withdrawal functionality
- Balance checking
- Account information display
- Secure account access using account number

Code submission (Github link)

https://github.com/shiyam2301/upskillCampus

Report submission (Github link) :

https://github.com/shiyam2301/upskillCampus

5 Proposed Design/ Model

The system is designed using **Object-Oriented Programming (OOP)** concepts.

Main components of the system:

1. **BankAccount Class**
    - Stores account details
    - Performs deposit and withdrawal operations
2. **BankSystem Class**
    - Manages multiple accounts
    - Handles account creation and login
3. **Main Class**
    - Provides the user interface
    - Allows users to perform banking operations

6 Performance Test

The system was tested with different banking operations to verify its functionality.

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Result
Create Account	Name, Account Number	Account created successfully
Deposit Money	Amount	Balance updated
Withdraw Money	Amount	Balance reduced
Check Balance	Account number	Balance displayed

6.2 Performance Outcome

The system successfully performed all basic banking operations without errors. The operations such as deposit, withdrawal, and balance inquiry worked as expected.

The program demonstrated the practical implementation of Java programming concepts.

7 My learnings

During this internship, I gained valuable experience in software development and programming. Some key learnings include:

- Understanding object-oriented programming concepts
- Developing console-based applications in Java
- Implementing real-world problem-solving using programming
- Improving logical thinking and debugging skills
- Learning about software design and system architecture

This internship has enhanced my technical skills and prepared me for future projects and professional work.

8 Future work scope

Although the current system performs basic banking operations, several improvements can be made in the future:

- Adding graphical user interface (GUI)
- Implementing database integration
- Adding secure authentication systems
- Developing a web-based banking platform
- Implementing transaction history and reports

These improvements would make the system more practical and closer to real-world banking applications.

9 . Screenshots:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\BankingInformationSystem> java Main
1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
1
Amount: 50000
Deposit Successful!

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
2
Amount: 1500
Withdrawal Successful!

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
3
Current Balance: 198500.0

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\BankingInformationSystem> java Main
3 Balance
4 Account Details
5 Logout
4
Account Holder: sivu
Account Number: 102
Balance: 150000.0

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
1
Amount: 50000
Deposit Successful!

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
2
Amount: 1500
Withdrawal Successful!

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\BankingInformationSystem> java Main
3 Exit
2
Enter Account Number: 102

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
4
Account Holder: sivu
Account Number: 102
Balance: 150000.0

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
1
Amount: 50000
Deposit Successful!

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\BankingInformationSystem> java Main
Enter Initial Balance: 150000
Account Created Successfully!

--- BANKING SYSTEM ---
1 Create Account
2 Login
3 Exit
2
Enter Account Number: 102

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
4
Account Holder: sivu
Account Number: 102
Balance: 150000.0

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\BankingInformationSystem> javac *.java
PS E:\BankingInformationSystem> java Main

--- BANKING SYSTEM ---
1 Create Account
2 Login
3 Exit
1
Enter Name: sivu
Enter Account Number: 102
Enter Initial Balance: 150000
Account Created Successfully!

--- BANKING SYSTEM ---
1 Create Account
2 Login
3 Exit
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\BankingInformationSystem> javac *.java
PS E:\BankingInformationSystem> java Main

 --- BANKING SYSTEM ---
 1 Create Account
 2 Login
 3 Exit
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\BankingInformationSystem> java Main
Deposit Successful!

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
2
Amount: 1500
Withdrawal Successful!

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
3
Current Balance: 198500.0

1 Deposit
2 Withdraw
3 Balance
4 Account Details
5 Logout
5

--- BANKING SYSTEM ---
1 Create Account
2 Login
3 Exit
3
PS E:\BankingInformationSystem>
```